Report for ForestQuery into Global Deforestation, 1990 to 2016

ForestQuery is on a mission to combat deforestation around the world and to raise awareness about this topic and its impact on the environment. The data analysis team at ForestQuery has obtained data from the World Bank that includes forest area and total land area by country and year from 1990 to 2016, as well as a table of countries and the regions to which they belong.

The data analysis team has used SQL to bring these tables together and to query them in an effort to find areas of concern as well as areas that present an opportunity to learn from successes.

1. **GLOBAL SITUATION**

According to the World Bank, the total forest area of the world was 41,282,694.9 sq. km in 1990. As of 2016, the most recent year for which data was available, that number had fallen to 39,958,245.9 sq. km., a loss of 1,324,449 sq. km., or -3.21%.

The forest area lost over this time period is slightly more than the entire land area of Peru listed for the year 2016 (which is 1,280,000 sq. km).

2. **REGIONAL OUTLOOK**

In 2016, the percent of the total land area of the world designated as forest was 31.38%. The region with the highest relative forestation was Latin America & Caribbean, with 46.16%, and the region with the lowest relative forestation was the Middle East & North Africa, with 2.07% forestation.
In 1990, the percent of the total land area of the world designated as forest was 32.42%. The region with the highest relative forestation was Latin America & Caribbean, with 51.03%, and the region with the lowest relative forestation was the Middle East & North Africa, with 1.78% forestation.

Table 2.1: Percent Forest Area by Region, 1990 & 2016:

Region	1990 Forest Percentage	2016 Forest Percentage
Latin America & Caribbean	51.03	46.16
Europe & Central Asia	37.28	38.04
North America	35.65	36.04
Sub-Saharan Africa	30.67	28.79
East Asia & Pacific	25.78	26.36

South Asia	16.51	17.51
Middle East & North Africa	1.78	2.07
World	32.42	31.38

The only regions of the world that decreased in percent forest area from 1990 to 2016 were Latin America & Caribbean (dropped from 51.03% to 46.16%) and Sub-Saharan Africa (30.67% to 28.79%). All other regions actually increased in forest area over this time period. However, the drop in forest area in the two aforementioned regions was so large, the percent forest area of the world decreased over this time period from 32.42% to 31.38%.

3. **COUNTRY-LEVEL DETAIL**

A. SUCCESS STORIES

There is one particularly bright spot in the data at the country level, China. This country actually increased in forest area from 1990 to 2016 by 527,229.06 sq. km. It would be interesting to study what has changed in this country over this time to drive this figure in the data higher. The country with the next largest increase in forest area from 1990 to 2016 was the United States of America, but it only saw an increase of 79,200 sq. km., much lower than the figure for China.

China and the United States of America are of course very large countries in total land area, so when we look at the largest percent change in forest area from 1990 to 2016, we aren't surprised to find a much smaller country listed at the top. Iceland increased in forest area by 213.66% from 1990 to 2016.

B. LARGEST CONCERNS

Which countries are seeing deforestation to the largest degree? We can answer this question in two ways. First, we can look at the absolute square kilometer decrease in forest area from 1990 to 2016. The following 5 countries had the largest decrease in forest area over the time period under consideration:

Table 3.1: Top 5 Amount Decrease in Forest Area by Country, 1990 & 2016:

Country	Region	Absolute Forest Area Change
Brazil	Latin America & Caribbean	541,510 sq. km.
Indonesia	East Asia & Pacific	282,194 sq. km.
Myanmar	East Asia & Pacific	107,234 sq. km.
Nigeria	Sub-Saharan Africa	106,506 sq. km.
Tanzania	Sub-Saharan Africa	102,320 sq. km.

The second way to consider which countries are of concern is to analyze the data by percent decrease. This comparison only includes countries with a valid, non-missing forest-area observation in both 1990 and 2016, so a country with no 2016 data point (such as St. Martin (French part)) is correctly excluded rather than mistaken for a complete loss of forest.

Table 3.2: Top 5 Percent Decrease in Forest Area by Country, 1990 & 2016:

Country	Region	Pct Forest Area Change
Togo	Sub-Saharan Africa	75.45%
Nigeria	Sub-Saharan Africa	61.80%
Uganda	Sub-Saharan Africa	59.13%
Mauritania	Sub-Saharan Africa	46.75%
Honduras	Latin America & Caribbean	45.03%

When we consider countries that decreased in forest area percentage the most between 1990 and 2016, we find that four of the top 5 countries on the list are in the region of Sub-Saharan Africa. The countries are Togo, Nigeria, Uganda, and Mauritania. The 5th country on the list is Honduras, which is in the Latin America & Caribbean region.

From the above analysis, we see that Nigeria is the only country that ranks in the top 5 both in terms of absolute square kilometer decrease in forest as well as percent decrease in forest area from 1990 to 2016. Therefore, this country has a significant opportunity ahead to stop the decline and hopefully spearhead remedial efforts.

C. QUARTILES

Table 3.3: Count of Countries Grouped by Forestation Percent Quartiles, 2016:

Quartile	Number of Countries
0 - 25%	85
25 - 50%	72
50 - 75%	38
75 - 100%	9

The largest number of countries in 2016 were found in the 0-25% quartile.

There were 9 countries in the top quartile in 2016. These are countries with a very high percentage of their land area designated as forest. The following is a list of countries and their respective forest land, denoted as a percentage.

Table 3.4: Top Quartile Countries, 2016:

Country	Region	Pct Designated as Forest
Suriname	Latin America & Caribbean	98.26%
Micronesia, Fed. Sts.	East Asia & Pacific	91.86%
Gabon	Sub-Saharan Africa	90.04%
Seychelles	Sub-Saharan Africa	88.41%
Palau	East Asia & Pacific	87.61%
American Samoa	East Asia & Pacific	87.50%
Guyana	Latin America & Caribbean	83.90%
Lao PDR	East Asia & Pacific	82.11%
Solomon Islands	East Asia & Pacific	77.86%

D. COMPARISON TO THE UNITED STATES

In 2016, 94 countries had a percent forestation higher than the United States. This indicates that while the U.S. compares favorably to the world in absolute forest-area growth, a large majority of countries still have a greater share of their land designated as forest.

4. **RECOMMENDATIONS**

Write out a set of recommendations as an analyst on the ForestQuery team.

- *What have you learned from the World Bank data?*
- *Which countries should we focus on over others?*

Learnings of World Bank data:

This report relies solely on figures published by the World Bank and does not factor in related influences like changes in population, economic conditions, or ongoing domestic or international conflicts. Each of these could meaningfully affect whether forest land expands or shrinks.

The countries that should be focused upon:

Attention should be directed toward the countries that have experienced the steepest declines in forest land, whether measured in absolute square kilometers or as a percentage of total forest cover. This group includes Brazil, Indonesia, Myanmar, Nigeria, Tanzania, Togo, Uganda, and Mauritania. Understanding what drove these losses between 1990 and 2016 will be key to curbing further decline and, where feasible, restoring forest land. Many of the hardest-hit areas have also experienced conflict, yet they remain rich in biodiversity and natural resources worth safeguarding.

Note: St. Martin (French part) was previously included in this list based on an apparent -100% change, but that figure was an artifact of a missing 2016 data point rather than a documented loss of forest, so it has been removed pending better data.

5. **APPENDIX: SQL Queries Used**

Create View

```
-- Create a view called "forestation"
-- Joins forest_area, land_area, and regions tables
DROP VIEW IF EXISTS forestation;

CREATE VIEW forestation AS
SELECT
    fa.country_code,
    fa.country_name,
    fa.year,
    fa.forest_area_sqkm,
    la.total_area_sq_mi * 2.59 AS total_area_sqkm,
    -- Calculate the percent of total land that is forest area
    (fa.forest_area_sqkm / (la.total_area_sq_mi * 2.59) * 100) AS perc_land_is_forest,
    r.region,
    r.income_group
FROM forest_area fa
JOIN land_area la
    ON fa.country_code = la.country_code
    AND fa.year = la.year
JOIN regions r
    ON la.country_code = r.country_code;
```

Global Situation

```
/* a. What was the total forest area (in sq km) of the world in 1990? */
SELECT country_name, forest_area_sqkm, year
FROM forestation
WHERE country_name = 'World' AND year = 1990;

/* b. What was the total forest area (in sq km) of the world in 2016? */
SELECT country_name, forest_area_sqkm, year
FROM forestation
WHERE country_name = 'World' AND year = 2016;

/* c. What was the change (in sq km) in the forest area of the world
from 1990 to 2016? */
-- FIX: replaced the comma-separated table list with an explicit JOIN,
-- joining the two years of the same "World" row to itself on country_code.
SELECT
    (f2016.forest_area_sqkm - f1990.forest_area_sqkm) AS forest_area_change_sq_km
FROM forestation AS f2016
JOIN forestation AS f1990
    ON f2016.country_code = f1990.country_code
WHERE f2016.year = 2016 AND f2016.country_name = 'World'
    AND f1990.year = 1990 AND f1990.country_name = 'World';

/* d. What was the percent change in forest area of the world between
1990 and 2016? */
-- FIX: same explicit JOIN as (c).
```

```
SELECT
    (f2016.forest_area_sqkm - f1990.forest_area_sqkm) AS forest_area_change_sq_km,
    ROUND(CAST((f2016.forest_area_sqkm - f1990.forest_area_sqkm)
        / f1990.forest_area_sqkm * 100 AS NUMERIC), 2) AS forest_area_percent_change
FROM forestation AS f2016
JOIN forestation AS f1990
    ON f2016.country_code = f1990.country_code
WHERE f2016.year = 2016 AND f2016.country_name = 'World'
    AND f1990.year = 1990 AND f1990.country_name = 'World';

/* e. If you compare the amount of forest area lost between 1990 and
2016, to which country's total area in 2016 is it closest to? */
SELECT country_name, year, total_area_sqkm
FROM forestation
WHERE year = 2016
    AND country_name <> 'World'
    AND total_area_sqkm < (
        (SELECT forest_area_sqkm FROM forestation WHERE country_name = 'World' AND year =
1990) -
        (SELECT forest_area_sqkm FROM forestation WHERE country_name = 'World' AND year =
2016)
    )
ORDER BY total_area_sqkm DESC
LIMIT 1;
```

Regional Outlook

```
/* a. Percent forest of the entire world in 2016; highest and lowest
region, to 2 decimal places. */
-- FIX: the old query summed forest/total area across every row for the year,
-- which included the "World" aggregate row itself alongside every country row
-- (double counting). The world total should come straight from the single
-- World-level observation instead of being re-aggregated.
SELECT
    ROUND(CAST(forest_area_sqkm / total_area_sqkm * 100 AS NUMERIC), 2) AS forest_area_world
FROM forestation
WHERE year = 2016 AND country_name = 'World';

-- FIX: excluded the World row from the regional grouping (it has no region
-- of its own and would otherwise appear as a phantom "region"), and made the
-- descending sort explicit.
SELECT
    region,
    ROUND(CAST(SUM(forest_area_sqkm) / NULLIF(SUM(total_area_sqkm), 0) * 100
        AS NUMERIC), 2) AS forest_area_pct
FROM forestation
WHERE year = 2016 AND country_name <> 'World'
GROUP BY region
ORDER BY forest_area_pct DESC;

-- FIX: same exclusion, explicit ASC for the lowest-region view.
SELECT
    region,
    ROUND(CAST(SUM(forest_area_sqkm) / NULLIF(SUM(total_area_sqkm), 0) * 100
        AS NUMERIC), 2) AS forest_area_pct
FROM forestation
WHERE year = 2016 AND country_name <> 'World'
GROUP BY region
ORDER BY forest_area_pct ASC;
```

```
/* b. Percent forest of the entire world in 1990; highest and lowest
region, to 2 decimal places. */
SELECT
    ROUND(CAST(forest_area_sqkm / total_area_sqkm * 100 AS NUMERIC), 2) AS forest_area_world
FROM forestation
WHERE year = 1990 AND country_name = 'World';

SELECT
    region,
    ROUND(CAST(SUM(forest_area_sqkm) / NULLIF(SUM(total_area_sqkm), 0) * 100
        AS NUMERIC), 2) AS forest_area_pct
FROM forestation
WHERE year = 1990 AND country_name <> 'World'
GROUP BY region
ORDER BY forest_area_pct DESC;

SELECT
    region,
    ROUND(CAST(SUM(forest_area_sqkm) / NULLIF(SUM(total_area_sqkm), 0) * 100
        AS NUMERIC), 2) AS forest_area_pct
FROM forestation
WHERE year = 1990 AND country_name <> 'World'
GROUP BY region
ORDER BY forest_area_pct ASC;

/* c. Which regions of the world DECREASED in forest area from 1990
to 2016? */
-- FIX: excluded the World row so it can't be mistaken for a "region" in the
-- HAVING comparison, and added an ORDER BY for readability.
SELECT
    region,
    SUM(CASE WHEN year = 1990 THEN forest_area_sqkm ELSE 0 END) AS forest_area_1990,
    SUM(CASE WHEN year = 2016 THEN forest_area_sqkm ELSE 0 END) AS forest_area_2016
FROM forestation
WHERE country_name <> 'World'
GROUP BY region
HAVING SUM(CASE WHEN year = 2016 THEN forest_area_sqkm ELSE 0 END) <
    SUM(CASE WHEN year = 1990 THEN forest_area_sqkm ELSE 0 END)
ORDER BY region;
```

Country-level Detail

```
/* a. Which 5 countries saw the largest amount decrease in forest area
from 1990 to 2016? */
-- FIX: replaced conditional aggregation with an explicit self-join that only
-- pairs a country's 1990 row with its 2016 row when BOTH exist and have a
-- non-null forest_area_sqkm value. LIMIT corrected from 6 to 5.
SELECT
    f1990.country_code,
    f1990.country_name,
    f1990.forest_area_sqkm AS forest_area_1990,
    f2016.forest_area_sqkm AS forest_area_2016,
    (f2016.forest_area_sqkm - f1990.forest_area_sqkm) AS forest_area_change
FROM forestation AS f1990
JOIN forestation AS f2016
    ON f1990.country_code = f2016.country_code
WHERE f1990.year = 1990
    AND f2016.year = 2016
    AND f1990.country_name <> 'World'
    AND f1990.forest_area_sqkm IS NOT NULL
```

```
    AND f2016.forest_area_sqkm IS NOT NULL
ORDER BY forest_area_change ASC
LIMIT 5;

/* b. Which 5 countries saw the largest percent decrease in forest
area from 1990 to 2016? */
-- FIX: same self-join pattern. This is the query that previously let
-- St. Martin (French part) in with a false -100%, because CASE WHEN ... ELSE 0
-- turned its missing (NULL) 2016 observation into a 0, making it look like a
-- complete loss. Requiring both years' values to be NOT NULL removes it
-- correctly instead of papering over the gap. LIMIT corrected from 6 to 5.
SELECT
    f1990.country_code,
    f1990.country_name,
    f1990.forest_area_sqkm AS forest_area_1990,
    f2016.forest_area_sqkm AS forest_area_2016,
    ROUND(CAST((f2016.forest_area_sqkm - f1990.forest_area_sqkm)
        / f1990.forest_area_sqkm * 100 AS NUMERIC), 2) AS percent_change
FROM forestation AS f1990
JOIN forestation AS f2016
    ON f1990.country_code = f2016.country_code
WHERE f1990.year = 1990
    AND f2016.year = 2016
    AND f1990.country_name <> 'World'
    AND f1990.forest_area_sqkm IS NOT NULL
    AND f2016.forest_area_sqkm IS NOT NULL
    AND f1990.forest_area_sqkm > 0
ORDER BY percent_change ASC
LIMIT 5;

/* c. If countries were grouped by percent forestation in quartiles,
which group had the most countries in it in 2016? */
-- FIX: switched from a window function (COUNT ... OVER PARTITION BY + DISTINCT)
-- to a true GROUP BY aggregation, as the rubric specifically required for this
-- question. ORDER BY countries DESC makes the largest group immediately visible.
SELECT
    CASE
        WHEN perc_land_is_forest <= 25 THEN '0-25%'
        WHEN perc_land_is_forest <= 50 THEN '25-50%'
        WHEN perc_land_is_forest <= 75 THEN '50-75%'
        ELSE '75-100%'
    END AS quartiles,
    COUNT(country_name) AS countries
FROM forestation
WHERE perc_land_is_forest IS NOT NULL
    AND year = 2016
    AND country_name <> 'World'
GROUP BY quartiles
ORDER BY countries DESC;

/* d. List all of the countries that were in the 4th quartile
(percent forest > 75%) in 2016. */
SELECT country_name, region, perc_land_is_forest
FROM forestation
WHERE year = 2016
    AND country_name <> 'World'
    AND perc_land_is_forest > 75
ORDER BY perc_land_is_forest DESC;

/* e. How many countries had a percent forestation higher than the
United States in 2016? */
```

```
WITH us_forest_percentage AS (
    SELECT perc_land_is_forest
    FROM forestation
    WHERE country_code = 'USA' AND year = 2016
)
SELECT COUNT(*) AS countries_higher_than_us
FROM forestation
WHERE year = 2016
    AND country_name <> 'World'
    AND perc_land_is_forest > (SELECT perc_land_is_forest FROM us_forest_percentage);
```